

DynamiX Device Reference

Every device, every knob, what it means

Generated from the live device registry

September 22, 2026 · kernel figures drawn by `make_figures.py` from the app’s own kernel code

How to read this document

A *device* is one processing tool in a layer’s chain. **Transforms** compute (they head chains and produce a result bundle); **filters** select from a computed result (list indexes and masks — scrubbing a filter knob redraws without recomputing). Chains are always transforms-then-filters; interleaving is rejected at construction.

Conventions that hold everywhere:

- **Scales are stored in pixels**, converted only for display.
- **The wavelet family is a property of the method**, never an orthogonal knob: each tool’s menu lists only the kernel class its method admits, under the method’s own names.
- **Inert knobs**: a parameter that binds only under one choice of another parameter (e.g. q only for q-family wavelets) is simply ignored otherwise.
- **Orientations are (angle, frame)** with `NorthRef` $\in \{\text{grid, true, magnetic}\}$ — never a bare float.
- **No-support masking**: pixels with no honest signal support (nodata flats, FFT dust) are NaN in every exponent map — absence, never fabrication.
- Each table lists: parameter (UI label), default, hard range, and meaning. Soft ranges (knob-drag limits) are narrower than hard ranges.

Wavelet normalization: L^1 , not L^2

Every wavelet in the app is normalized in the L^1 (pure-dilation) convention, $\psi_a(x) = a^{-d} \psi(x/a)$ in $d = 2$ dimensions, so that at a Hölder singularity the transform scales as $|T_\psi[f](x, a)| \sim a^{h(x)}$ with no offset. The L^2 convention ($a^{-d/2}$, unit energy $\int \psi^2 = 1$ — e.g. the normalization constants A and A_q of Borges et al. 2004, whose transform carries $a^{-1/2}$) shifts every exponent by $d/2$. Turiel’s microcanonical formalism uses the same L^1 dilation $\Psi_r(x) = r^{-d} \Psi(x/r)$ (Turiel et al. 2008, eqs. 6–8). Borges et al.’s own WTMM example agrees in practice: their eq. 41 reads $T \sim a^\alpha$ (their fig. 5 recovers $\alpha = 0.5$), which holds for L^1 scaling, not for the $a^{-1/2}$ of their eq. 10.

- **wtmm2d normalizes in the Fourier domain**. Each filter is built as $\hat{\psi}(a\mathbf{k})$ from a smoothing function with $\hat{\theta}(0) = 1$ (unit mass at every scale), times the derivative factor $i a\mathbf{k}$: scale enters only through $a\mathbf{k}$, which *is* the L^1 dilation.
- **The Hölder tools normalize in real space**, on the pixel grid: positive kernels (`holder_measure`) to unit discrete mass; Mexican hats (`holder_multiaffine`) to exact zero mean, then unit L^1 norm ($\sum |\psi_r| = 1$), at every scale.
- A constant prefactor that does not depend on the scale (such as A or A_q) cannot change any exponent — it only shifts $\log |T|$ — and the app’s own normalization replaces it anyway. Only the *scale dependence* of the normalization matters.

- **Validated on a synthetic cusp** $|x - x_0|^h$: both routes recover the slope h (tests pin it; an L^2 slip would read $h + 1$ in 2-D). The smallest scales are biased by the discretization of both kernel and signal: at a sampled cusp the slope settles above $\sigma \approx 5$ px (`wtmm2d`) and $r \approx 10$ px (the Mexican hats), later for sharp-cutoff kernels ($q_T < 1$).
- **Two different q 's.** q_T (the knob *Tsallis* q of the Hölder tools) is the Tsallis/Borges shape parameter of a kernel; q in $Z(q, a)$, $\tau(q)$ and $D(h)$ is the moment order of the partition functions. They are unrelated.

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1 Field preparation

1.1 `noise` — dither / perturbation

Adds controlled noise to the field before analysis. Its main use is *dithering* quantized data ($\pm \frac{1}{2}$ LSB) so that quantization plateaus do not read as zero-gradient flats.

Parameter	Default	Range	Meaning
<code>dist</code> (Distribution)	uniform	uniform / gaussian	noise law
<code>amplitude</code>	0	$[0, 10^9]$	0 = auto $\frac{1}{2}$ LSB; uniform: half-width; gaussian: σ
<code>seed</code>	0	$[0, 2^{31}]$	RNG seed (deterministic reruns)

2 Primary analyzers

Each primary analyzer is its own representation of the field. Dropping a *different* primary analyzer onto an analyzed layer forks a sibling layer rather than mutating the chain; consumers (filters) follow their producer on forks.

2.1 wtmm2d — canonical WTMM (Arnéodo)

The canonical multifractal machinery: a continuous wavelet transform over a geometric scale grid, modulus *maxima* detected per scale (along-gradient NMS), maxima chained across scales into maxima lines, feeding partition functions $\rightarrow \tau(q) \rightarrow$ Legendre $D(h)$. The maxima are dense by design — they are the statistical support, not a lineament map (use the filters, or the other analyzers, for sparse features). Legendre spectra are concave hulls by construction; the microcanonical tools (§4) measure $D(h)$ directly — together they diagnose non-concavity.

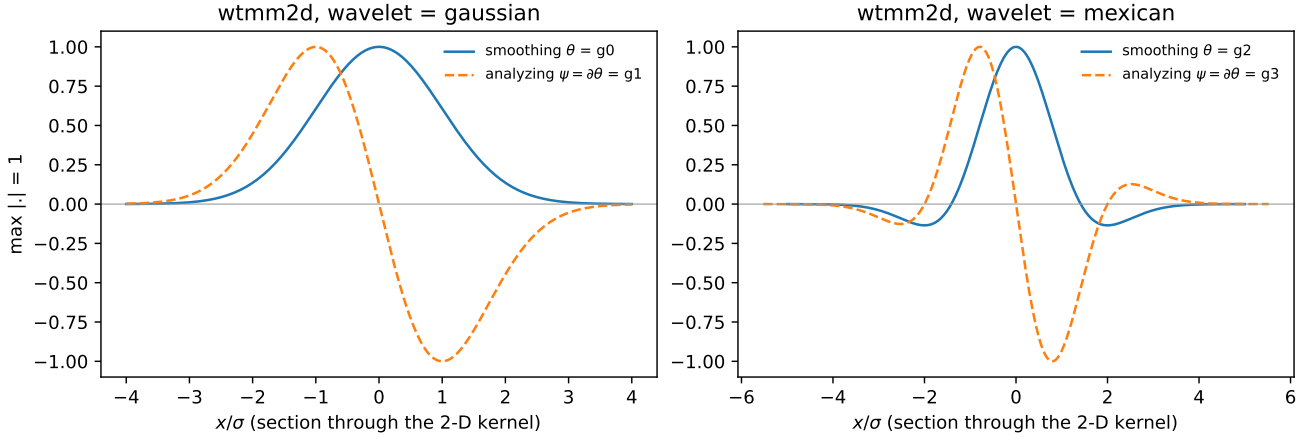


Figure 1: **wtmm2d**: the smoothing function θ and the analyzing wavelet $\psi = \partial\theta$ on the same axes (sections through the 2-D kernels, x in units of the smoothing σ). The wavelet is the derivative of the smoothing function, so the modulus maxima of ψ are the edges of θ -smoothed data.

Parameter	Default	Range	Meaning
n_oct (Octaves)	3	[1, 8]	scale range: a spans n_oct octaves above $a_{\min}$
n_voice	4	[1, 16]	scales per octave (grid density)
a_min ($a_{\min}$)	1.0	[0.25, 16] px	finest analyzed scale
wavelet	mexican	mexican / gaussian	CWT kernel (Mexican hat = LoG; gaussian = gradient pair)
dither	off	bool	$\pm \frac{1}{2}$ LSB dither for quantized rasters
fracint_alpha (η)	1.0	[-2, 3]	per-scale a^η lift (pseudo fractional integration; shifts effective h by η)
min_chain_len	2	[2, 64]	discard maxima lines spanning fewer scales
smooth	on	bool	smooth maxima-line traces
thresh	0.001	[0, 1]	modulus floor for maxima detection (fraction)
dist2_max	50	[1, 500] px ²	max squared jump when chaining maxima across scales
box_ratio	1.0	[0.1, 8]	chaining search-box anisotropy
similitude	0.8	[0, 1]	cross-scale similarity acceptance for chaining

2.2 wtmm2d.roi — WTMM over one ROI window

The parent’s WTMM re-pointed at a native-resolution window: same knobs as `wtmm2d` (minus the fractional-integration lift) plus the window itself. Used by the overview drill-down: draw an ROI on the decimated whole-extent, compute native.

Parameter	Default	Range	Meaning
<code>roi_row, roi_col</code>	0	px	window origin (native pixels)
<code>roi_h, roi_w</code>	512	≥ 8 px	window size
<code>boundary</code>	auto	auto / reflective	boundary handling at the window edge

2.3 mz_edges — Mallat–Zhong dyadic multiscale edges

“Discrete Approximation of Multiscale Edges” (Mallat–Zhong 1992) as a peer analyzer: the à-trous dyadic transform on the mirrored torus, per-level gradient-pair maxima (the multiscale Canny), wrap-merged into per-scale H-lines (`line_id`). Stored moduli are the paper’s own λ -normalized values (Table II). Scales are dyadic, 2^{k+1} px. The `frac_bspline` wavelet fractionalizes the refinement order (Unser–Blu): smoothness order α becomes real, one vanishing moment always (maxima-are-edges *is* the method); $\alpha = 3$ is the paper’s wavelet exactly, with numeric λ_j^α from the rediscovered Table-II definition (discrete step-edge peak over $\theta_\alpha(0)$). Reconstruction preview requires `mz_spline` (fractional POCS is a recorded follow-up).

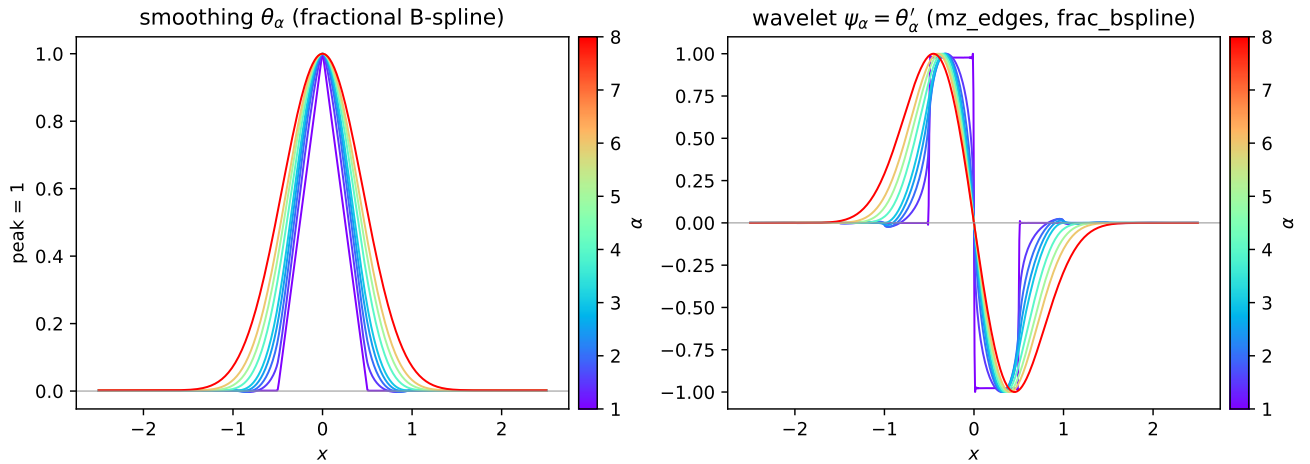


Figure 2: `mz_edges, frac_bspline`: the smoothing function $\theta_\alpha(x) = 2\beta_\alpha^*(2x)$, the *symmetric* fractional B-spline of degree α (Unser & Blu 1999, eqs. 10–11; $\hat{\beta}_\alpha^*(\omega) = |\sin(\omega/2)/(\omega/2)|^{\alpha+1}$), and the wavelet $\psi_\alpha = \theta'_\alpha$ as α varies — the analytic limit the filter cascade is pinned to (its maxima sit exactly where the classical $\alpha = 3$ path puts them: no causal phase). α is θ ’s Hölder degree; its approximation order is $\alpha + 1$. $\alpha = 1$: a triangle and a box pair; $\alpha = 3$: the paper’s quadratic-spline wavelet. Only odd integer α gives a compactly supported classical B-spline; other orders decay algebraically, and as $\alpha \rightarrow 0$ the symmetric spline is *not* a box but develops logarithmic peaks and oscillating tails (Unser & Blu, fig. 2). This is Mallat–Zhong’s dyadic *derivative* wavelet built on a fractional spline — exactly one vanishing moment ($\hat{\psi} \sim i\omega$ at low frequency), since maxima-are-edges is the method — not one of Unser & Blu’s fractional spline wavelet bases, which behave like fractional differentiators $|\omega|^{\alpha+1}$.

Parameter	Default	Range	Meaning
<code>n_levels</code> (Levels J)	4	[1, 10]	dyadic levels; scales 2, 4, $\dots$, 2^J px
<code>coarse</code>	full	full / thumbnail	coarse-channel policy (mapping vs coding mode)
<code>dither</code>	off	bool	$\pm \frac{1}{2}$ LSB dither before analysis
<code>wavelet</code>	<code>mz_spline</code>	<code>mz_spline</code> / <code>frac_bspline</code>	the paper’s quadratic-spline bank, or the fractional order

alpha (α)	3.0	[0.1, 8]	fractional spline order (binds under frac_bspline; 3 = the paper)
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2.4 wavelet_skeleton — medial-axis (modulus-minima) extractor

Tang–You 2003 (the constructed compact-support wavelet κ : support radius = scale, maxima-pair separation = scale, width- and gray-invariant) + You et al. 2006 (modulus *minima* sit on the medial axis, scale-independently). A detector, not statistics: directional minima gated by a flanking ridge *pair* (each side within reach $\sim s$ must clear **edge_frac** of the peak modulus), valley contrast (**t_frac** of the flanking ridge — the paper’s threshold T made local), and pair dominance (a pair under **pair_dom** of the strongest reachable ridge is someone else’s slope). Border and nodata neighborhoods (one kernel support) are distrusted. On **input=gradient** the detector runs over $\|\nabla s\|$, where a diffused scarp (an *edge* in the signal) becomes a ribbon whose axis is the fault trace. Output is single-scale extrema (H-lines via **line_id**).

Parameter	Default	Range	Meaning
s1 (s_1)	6	[2, 64] px	stage-1 scale = kernel support radius; should cover the ribbon width
s2 (s_2)	6	[2, 64] px	refinement-stage scale
t_frac (T/ridge)	0.5	[0.1, 0.9]	keep minima below this fraction of their own flanking ridge
edge_frac (Edge/peak)	0.1	[0.01, 0.6]	flanking-pair significance vs the <i>peak</i> modulus; lower to admit faint features
n_stages	2	[1, 5]	WT passes (1 = detection only; each further pass re-runs on the skeleton image)
input wavelet	signal tang_you	signal / gradient tang_you / gaussian / bspline2	ribbons in the signal, or scarps via $\ \nabla s\ $ the papers’ own comparison set; theorems hold for κ only
pair_dom (Pair dom.)	0.5	[0, 0.9]	pair-dominance gate; 0 disables

2.5 cdf_edges — complex cross-diffusion edges

One complex cross-diffusion evolution (Gilboa small- θ ; Barbeiro 2023 for the scale-space reading), snapshotted at the dyadic schedule $n = \sigma^2/(2 \cos \theta)/dt$: the Re channel evolves as the Gaussian-smoothed image and $\text{Im}/\theta \approx t \cdot \text{LoG}$ (machine-pinned), so one evolution *is* a Mexican-hat scale space. Edge modes: **mz** = grad-Re NMS maxima (*our* graft — the M–Z definition applied to a different generator, making **cdf_edges** directly comparable to **mz_edges** at matched scales); **marr** = Im zero crossings (the CDF authors’ own edge concept — Barbeiro’s v “in the role of edge detector”). The nonlinear variant modulates diffusivity by the Im channel (Perona–Malik-style edge stopping through k). Final Re (**filtered**) and Im (**edge_channel**) rasters ride the result. Scales are effective Gaussian σ in px.

Parameter	Default	Range	Meaning
n_levels (Levels J)	4	[1, 8]	dyadic sigmas $1, 2, \dots, 2^{J-1}$ px; cost is quadratic in the top σ
variant k_edge (k)	linear 1.0	linear / nonlinear [0.05, 100]	LCDF, or NCDF with edge-stopping Perona–Malik edge threshold, image units (nonlinear only)
theta (θ)	0.1047	[0.02, 0.5] rad	complex-diffusion angle (small- θ regime; $\pi/30$ default)
dt edge_mode (Edges)	0.2 mz	[0.05, 0.24] mz / marr	explicit-Euler step (stability-capped) Canny/M–Z maxima of Re, or Marr zero crossings of Im
floor	0.05	[0.01, 0.5]	detection floor (fraction of the per-snapshot maximum)

2.6 pm_edges — Perona–Malik anisotropic diffusion edges

One Perona–Malik (1990) evolution — real, gradient-driven conduction $g(|\nabla I|)$, $\lambda \leq \frac{1}{4}$, adiabatic borders — snapshotted at the same nominal dyadic σ schedule as `cdf_edges`, with gradient-NMS maxima per snapshot. What it adds: past the flux peak $\phi(s) = s g(s)$ the diffusion runs locally *backward*, sharpening edges while the discrete maximum principle keeps it stable, so edges stay sharp and in place across the stack. Scales are nominal effective σ (the $K \rightarrow \infty$ heat-equation limit); `filtered` (the final diffused raster) rides the result.

Parameter	Default	Range	Meaning
<code>n_levels</code> (Levels J)	4	[1, 8]	dyadic nominal sigmas
<code>k_edge</code> (K)	1.0	[0.001, 1000]	edge threshold in image units: contrasts above K sharpen, below it blur
<code>g</code>	exp	exp / frac	conduction function (the paper’s two)
<code>lam</code> (λ)	0.2	[0.001, 0.25]	explicit step (stability needs $\lambda \leq \frac{1}{4}$)
<code>show</code>	edges	edges / filtered	canvas shows the edge stack, or the diffused raster
<code>detector</code>	nms	nms / follow	maxima detector
<code>interpolate</code>	off	bool	sub-pixel refinement of the maxima
<code>floor</code>	0.05	[0, 0.5]	detection floor (fraction of the per-snapshot maximum)

3 Decompositions

Standalone transforms heading their own chain, displaying through the ordinary raster path.

3.1 pca — principal components of a multi-band field

Principal component analysis over the bands of a multi-component field ($n_y, n_x, n_c \geq 2$) (covariance-trick PCA); shows one component as a raster.

Parameter	Default	Range	Meaning
<code>n_components</code> (Components)	6	[1, 64]	components computed
<code>component</code> (Show PC)	1	[1, 64]	the component shown
<code>standardize</code>	off	bool	scale each band to unit variance first

3.2 tucker_havok — Tucker (HOSVD) decomposition, optionally delay-embedded

A Tucker decomposition of the field, either of its own modes (rows, columns[, band]) or of a HAVOK-style time-delay (Hankel) embedding along one axis — built as a stride view, never materialized. Shows the rank-truncated reconstruction or its residual.

Parameter	Default	Range	Meaning
<code>embed</code> (Embedding)	delay	delay / none	Hankel delay embedding, or the array’s own modes
<code>axis</code> (Tape axis)	rows	rows / cols	the axis the delay embedding runs along
<code>n_delays</code> (Delays)	32	[2, 512]	delay-embedding depth
<code>rank_delay</code>	4	[0, 512]	Tucker rank of the delay mode (0 = full)
<code>rank_time, rank_space</code>	0	[0, 10^5]	ranks of the rows/cols modes (0 = full)
<code>rank_band</code>	0	[0, 64]	rank of the band mode (0 = full)
<code>sweeps</code> (HOOI sweeps)	0	[0, 16]	HOOI refinement sweeps after the HOSVD
<code>show</code>	recon	recon / residual	what the canvas shows

4 Microcanonical tools

Per-pixel singularity exponents $h(x)$ from the model $T(x, r) \approx \alpha(x) r^{h(x)}$ (Turiel 2008), $D(h)$ by direct histogram — the formalism that is *not* canonical/Legendre. One tool per functional; both expose the Pont 2006 estimator pair: **regression** (per-pixel log-log slope over the scale grid; the prefactor cancels exactly; resolves both limbs of $D(h)$) and **punctual** (the finest scale alone, ensemble-mean normalized; sharpest localization, the most-singular-set extractor; cannot resolve the right limb; carries an $\alpha(x)/\langle\alpha\rangle$ bias). Both estimators share the same cross-scale no-support mask — they agree on *where* an exponent exists and differ in *how*. Under punctual, $\kappa/\mathbf{n_scales}$ shape only the mask and **r2_map** is honestly all-NaN.

4.1 holder_measure — gradient-measure route

T = positive unit-mass kernel over $\|\nabla s\|$ (finite differences first, then g_0 -class smoothing). For measure-like / edge-sparse fields; slope = h . The scale knob r_1 is the kernel’s *dilation scale* (for the Gaussian: its standard deviation) — positive kernels have no zero crossings, and their resolution floor is the pixel. Kernel trade-off (Turiel App. A): Gaussian resolves every h ; Lorentzian truncates at $h < 2\beta - d$ but localizes sharpest; q-Gaussian interpolates; **frac_gaussian** fractionalizes the envelope, $e^{-\rho^n/2}$ ($n=2$ is the Gaussian exactly).

The q-Gaussian is Tsallis’s $e_q^{-\rho^2/2r^2}$ with $e_q^x = [1 + (1 - q)x]_+^{1/(1-q)}$ (Borges et al. 2004, eq. 9), for $-1 \leq q \leq 3$. Above $q = 1$ it has power-law tails (Student-t / Lorentzian: $q \equiv 1 + 1/\beta$ with ρ^2/r^2 rescaled); $q = 1$ is the Gaussian; below $q = 1$ it is *compactly supported*, radius $r\sqrt{2/(1-q)}$ — $q = 0$ is the Epanechnikov paraboloid, $q = -1$ a half-dome of radius r , and $q \rightarrow -\infty$ the Dirac δ . For $q \geq 2$ the 2-D kernel has no finite mass (the Lorentzian $\beta \leq 1$ case), so the result depends on the field size; the q knob shows a warning there.

4.2 holder_multiaffine — wavelet-projection route

$T = |\psi_r * s|$ with a zero-mean wavelet applied to the *raw signal* (derivative order lives inside the kernel). For smooth / function-class fields (DEMs); slope = γ , with $h_{\text{measure}} = \gamma - 1$. Here r_1 is the *zero-crossing radius*: every wavelet in the menu is calibrated to change sign at radius r , so the fig.-2 minimum-resolution rule reads $r_1 \geq 1$ uniformly. The g-ladder is the vanishing-moment axis (order n measures only $\gamma < n$): **g1/g3** are the fractional family at $n=1, 3$; **g2** is the Ricker; **lorentzian_marr** is the Laplacian (Marr operator) of the Lorentzian envelope (true tail $\rho^{-2\beta-2}$, truncation $\gamma \geq 2\beta$); **frac_gaussian** makes the order a real knob ($\psi \propto {}_1F_1(\frac{n+2}{2}; 1; -\rho^2/2\sigma^2)$, $n=2$ reproduces the Ricker exactly): n is a fractional *differentiation* order ($\hat{\psi} \sim |k|^n$ at low frequency), so exact moments vanish only for the integer orders below n — a fractional n is not “ n vanishing moments” — and $\gamma < n$ is measurable.

q_mexican is the q-Mexican hat of Borges et al. (2004, eq. 15): the second derivative of the q-Gaussian *raised to the power* $2 - q$. In 2-D (the Laplacian) this gives

$$\psi_q(\rho) \propto [1 - (2 - q)b\rho^2] [e_q^{-b\rho^2}]^q, \quad b = \frac{1}{(2 - q)r^2},$$

so the zero crossing sits at $\rho = r$ for every q , and $q = 1$ is the Ricker exactly. The paper’s $q < 3$ is the 1-D bound: in 2-D the Laplacian of $[e_q]^{2-q}$ vanishes identically at $q = 2$ and never changes sign above it, so the knob runs $-1 \leq q \leq 1.95$. For $q \geq 0$ the wavelet is bounded; for $q < 0$ it diverges at its cutoff (still integrable) and the knob shows a warning. Because $[e_q]^{2-q}$ is itself a q' -Gaussian with $q' = 1/(2 - q)$, this is the family of Laplacians of q-Gaussians under a relabelled index: **q_mexican** at q equals **lorentzian_marr** at $\beta = (2 - q)/(q - 1)$ for $1 < q < 2$ (e.g. $q = 1.5 \equiv \beta = 1$), with tail $\rho^{-2/(q-1)}$ and truncation $\gamma \geq 2(2 - q)/(q - 1)$.

The same r_1 value is not numerically comparable across the two tools, and γ values are not comparable across wavelets.

Parameter	Default	Range	Meaning (both holder tools)
estimator	regression	regression / punctual	Pont estimator pair (see above)
r_min (r_1)	1.0	[0.1, 32] px	finest scale (dilation scale / zero-crossing radius per tool)

kappa ($\kappa = r_2/r_1$)	8	[1.1, 200]	scale-range width; the knob that changes the physics
n_scales	6	[2, 64]	samples in the range (nearly irrelevant vs κ ; keep small)
wavelet	(per tool)	method-true menu	see tool text
q_tsallis (Tsallis q , q_T)	1.5	$[-1, 3]$ / $[-1, 1.95]$	q-family shape, measure / multiaffine (q-gaussian / q-mexican only); warns at measure $q \geq 2$, multiaffine $q < 0$
beta (β)	1.0	[0.1, 8]	Lorentzian tail exponent (Lorentzian family only)
frac_n (n)	2.0	[0.05, 8]	fractional order (frac-gaussian only)

4.3 band_recon_measure / band_recon_multiaffine — reconstruction from an h band

The same $h(x)$ engine as the matching holder tool (its params verbatim, shared by identity) + the singularity band $h \in [h_{lo}, h_{hi}]$ and an island sieve, then gradient-restricted Fourier reconstruction (Turiel eq. 65, symmetric-extension border fix). Sweeping the band and watching the reconstruction *is* the workflow; the Band dialog commits the variant matching the layer’s producer.

Parameter	Default	Range	Meaning (band knobs; engine knobs above)
h_lo / h_hi	-2 / 0	$[-6, 6]$	the band $h_{lo} \leq h < h_{hi}$, in the estimator’s own frame
min_island / max_island	0 / 0	px	drop mask components outside this size range before inversion
connectivity	8	8 / 4	island connectivity (8 keeps diagonal ribbons whole)

Superseded: **holder_map** and **band_recon** (the pre-split conflated tools, estimator $\times$ wavelet crossed) remain registered for saved projects and live in the browser’s Dev group; their physics is identical to the split tools at the old defaults.

5 Topology

5.1 chain_topology (Transform)

Builds the incidence graph over the WTMM maxima-line skeleton (no parameters). Must follow **wtmm2d**; downstream topology filters query the graph it builds.

5.2 min_vchains (Filter)

Keeps H-line points that at least **min_vchains** maxima lines (V-chains) thread. Default UI opens at 1 (show everything computed; turn up to prune).

6 Extrema filters

Act on the per-scale extrema layers of any producing analyzer (**wtmm2d**, **mz_edges**, **wavelet_skeleton**, **cdf_edges**).

Device.param	Default	Range	Meaning
scale_select.scale_idx	0	[0, 255]	pick ONE scale from the stack — <i>the scale scrubber</i> ; a list index, instant re
orientation_wedge.centre	0	deg	keep extrema whose gradient strike f wedge

<code>orientation.wedge.half_width</code>	90	[0, 90] deg	wedge half-width (90 = no-op)
<code>orientation.wedge.north</code>	grid	grid/true/magnetic	the NorthRef of centre ; frame error; multi-degree errors
<code>modulus.threshold.frac</code>	0	[0, 1]	per-point: keep $ W \geq \text{frac} \times \text{scale}$
<code>hline.length.min_len/max_len</code>	1 / 0	points	per-line point count range; kills orphan confetti and stitching seams; -1 orphan
<code>hline.modulus.frac</code>	0	[0, 1]	per- <i>line</i> modulus keep-fraction
<code>hline.modulus.mode</code>	sup	sup / sup_all_scales / mean / segments	line statistic: peak at this scale; peak threading chains (needs chains); mean per-segment between modulus minima

7 Chain filters (WTMM maxima lines)

Act on cross-scale chains — `wtmm2d` results only (the other analyzers stamp `chains:` []; their cross-scale linking is reserved work).

Device.param	Default	Range	Meaning
<code>chain_holder.cutoff</code>	-3	[-3, 3]	keep chains with Hölder $\geq$ cutoff
<code>chain_holder.estimator</code>	ols	ols / max	per-chain h : log-log OLS slope, or the max-modulus scaling read
<code>chain_modulus.threshold</code>	-99	[-99, 32]	keep chains whose max $\log_2 W \geq$ threshold
<code>chain_length.min_len</code>	2	[2, 64]	minimum scales spanned

7.1 chain_classify

Classifies chains by their Hölder exponent and geometry into step-like ($h \approx 0$: fault-scarp class) vs delta-like ($h \approx -1$: thin-ridge class) lineaments, with a straightness gate; flagged or excluded per **action**, excluded chains optionally drawn as ghosts.

Parameter	Default	Range	Meaning
<code>estimator</code>	ols	ols / max	per-chain h estimator
<code>fit_a_min</code>	2.0	[0.25, 16]	fit floor: scales below this are excluded from the h fit
<code>h_step_lo/hi</code>	-0.15 / 0.15	[-3, 3]	the step-class h window
<code>h_delta_lo/hi</code>	-1.2 / -0.8	[-3, 3]	the delta-class h window
<code>straightness_min</code>	0.85	[0, 1]	end-to-end / arc-length straightness gate
<code>action</code>	flag	flag / exclude	tag matches, or drop non-matches
<code>show_ghosts</code>	on	bool	draw excluded chains desaturated

8 Groups, points, dev

`group.paint` / `group.filter`: the arrangement view’s commit transaction — `group.paint` stamps committed group tags onto chain copies (`spec_json/signature` are written by the commit flow, not hand-edited); `group.filter` keeps or drops one named group.

`backproject`: registers a point catalogue onto a raster layer’s grid (the point-layer transform). `target` names the raster layer; the `_target_*` params are the resolved grid geometry, written by the shell — not hand-edited.

`stub.wavelet` / `stub.holder` / `stub.wedge`: parameter-schema test fixtures (the browser’s Dev group); no analysis role.

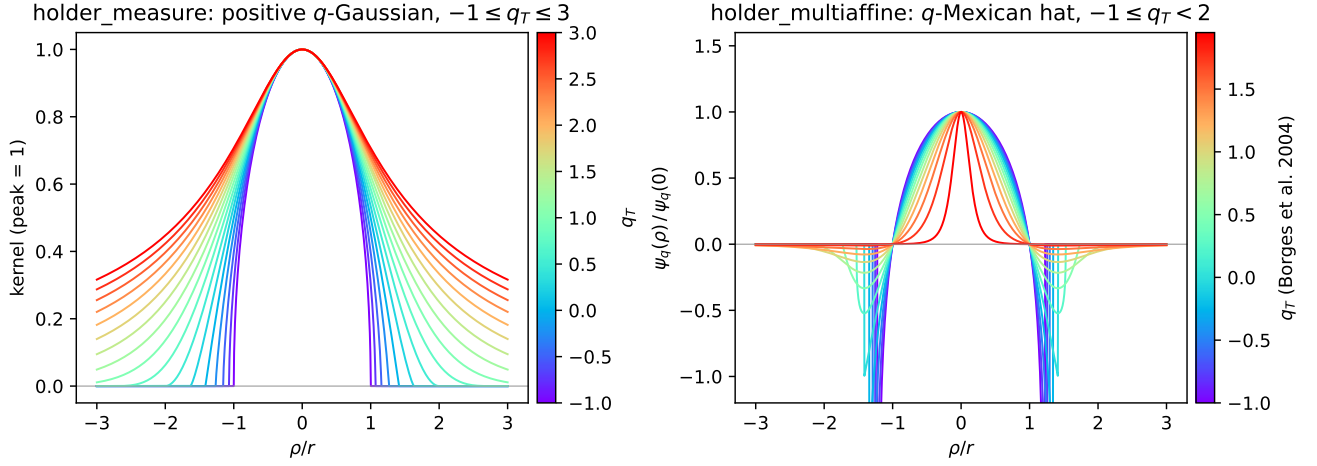


Figure 3: The q family. Left: `holder_measure`’s positive q -Gaussian, $-1 \leq q \leq 3$ (compact below $q = 1$). Right: `holder_multiaffine`’s q -Mexican hat (Borges et al. 2004, in 2-D), $-1 \leq q < 2$; every curve crosses zero at $\rho = r$, and for $q < 0$ the negative lobe diverges at the cutoff. Profiles scaled to 1 at the centre.

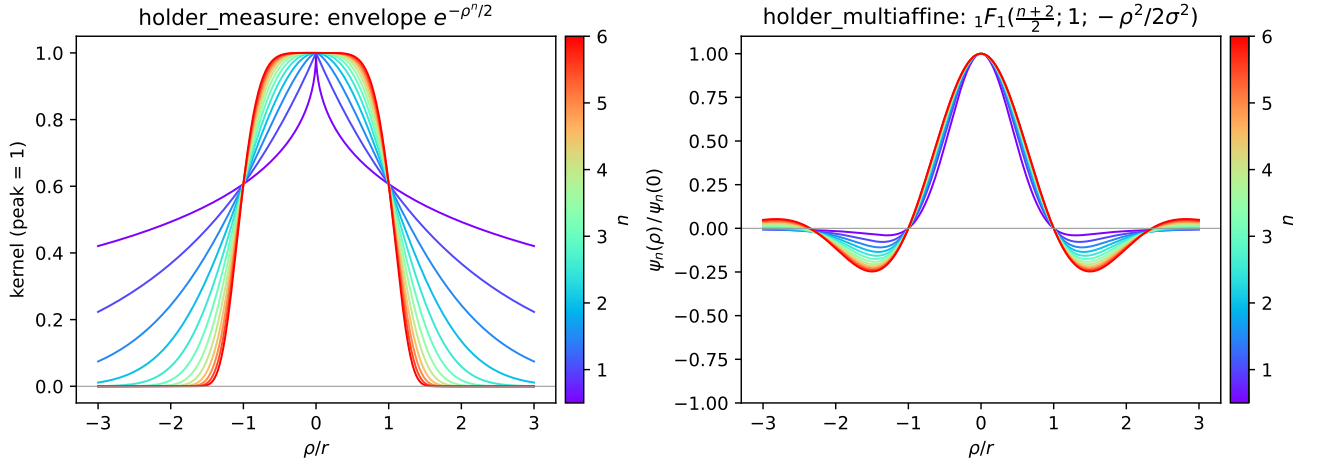


Figure 4: The fractional Gaussians. Left: `holder_measure`’s envelope $e^{-\rho^n/2}$ ($n < 2$ fattens the tail, $n > 2$ flattens the top; every curve passes $e^{-1/2}$ at $\rho = r$). Right: `holder_multiaffine`’s order- n wavelet ${}_1F_1(\frac{n+2}{2}; 1; -\rho^2/2\sigma^2)$, σ set per order so the first zero crossing sits at $\rho = r$.

9 Planned: q -generalized wavelets

Not implemented yet — recorded here so the design stays in one place. Borges et al. (2004) generalize the Mexican hat (above), the Morlet wavelet (a q -Gaussian-modulated plane wave, their eq. 23) and the trigonometric functions (an even/odd pair of q -trigonometric wavelets, their §4), and note that “other wavelets, as well as scaling functions, may be generalized along the lines here introduced, for instance, the Meyer wavelet or the sinc function.”

- **q -generalized g1 and g3 for wtmm2d.** WTMM’s analyzing wavelets must stay *derivatives of a smoothing function* ($\psi = \partial\theta$), so that maxima mark edges: the q versions are $\partial\theta_q$ with θ_q a q -Gaussian (g1) or its Mexican-hat (g3) — the same power- $(2 - q)$ recipe keeps closed forms, and the L^1 normalization is unchanged ($\hat{\theta}_q(0) = 1$).
- **q -Morlet** and the **q -trigonometric** pair (oriented/directional analysis).
- **q -Meyer** and **q -sinc**, as the paper suggests.

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Appendix: result contracts

- **Extrema schema** (all edge/skeleton analyzers): per-scale `{x, y, mod, arg, line_id}`; `line_id` groups H-lines, `-1` marks isolated points. `scales` is the matching scale array (px).
- **h-map contract** (holder tools): `h_map` displays in place of the field through the ordinary raster path; `r2_map` rides along (all-NaN under punctual). `raster_out` (band tools) displays the reconstruction; `h_map/band_mask` ride for the dialogs.
- **cdf extras**: `filtered` (final Re) and `edge_channel` (final Im) rasters.
- The vector (3-D) view draws the *finest* extrema layer only — put `scale_select` in the chain to choose which.